

Integrity Check (MIC)) as specified by [Section 3.6.1, “Generate and encrypt”](#). The secured outgoing message SHALL be:

A || C

4.8.3. Security Processing of Incoming Messages

All incoming message processing SHALL occur in a serialized manner. If an implementation chooses to process messages in a parallel manner, it must ensure that the behavior is opaque-box identical to a serialized processing implementation.

If the transport layer receives a secured message as indicated by the Session ID, it SHALL perform the following steps:

1. Determine the [Session Type](#), [Session ID](#), and [Message Counter](#) from the message header of the received message.
2. Obtain the Encryption Key associated with the [Session Context](#) of the given Session ID and Session Type:
 - a. If no key is found for the given Session ID, security processing SHALL indicate a failure to the next higher layer with a status of 'message security failed' and no further security processing SHALL be done on this message.
3. Execute the AEAD decryption and verification operation as specified in [Section 3.6.2, “Decrypt and verify”](#) with the following instantiations:
 - a. The bit string key *K* SHALL be the Encryption Key obtained from step 2;
 - b. The nonce *N* SHALL be the `CRYPTO_AEAD_NONCE_LENGTH_BYTES`-octet string constructed according to [Table 17, “Nonce”](#);
 - c. The parameter *C* SHALL be the encrypted and authenticated Message Payload;
 - d. The additional data octet string *A* SHALL be the authenticated Message Header;
 - e. `{success, P} = Crypto_AEAD_DecryptVerify(K, C, A, N)`
4. Return the result `{success, P}` of the AEAD operation:
 - a. If the `success` is `FALSE`, security processing SHALL fail and further processing SHALL NOT be performed on this message. An appropriate error SHOULD be raised to the upper layer to indicate the error.
 - b. Otherwise, set the octet string *PlaintextMessage* to the string

A || P

5. *PlaintextMessage* now represents the deciphered, authenticated, received message.
 - a. NOTE: The message has not yet undergone counter processing nor replay detection.
 - b. The *PlaintextMessage* SHALL be marked as successfully security processed and SHALL be released to the next processing layer.